

CAMBRIAN

Game Design Document and Development Report

CSCI 5499B/5999B — Project 1: Vertical Slice Prototype

Theme: Underwater

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1. Game Overview

Cambrian is a 2D underwater survival and evolution prototype built around a generational progression loop (yes its spores cell stage without customization and from a side view). The player controls a primitive proto-fish in a Cambrian era inspired ecosystem, feeds on plankton to earn Evolution Points (EP), searches for mates to preserve those points, avoids or fights predators, and eventually dies. Death ends the current generation and opens an evolution screen where preserved EP can be spent on inherited adaptations for the next generation.

The central design goal is to make evolution function as gameplay rather than as a detached upgrade menu of stats. Each adaptation changes how the player interacts with the environment: jaws enable combat, paired fins enable immediate directional changes, an improved tail increases mobility, sensory organs warn of nearby predators, and dermal armor improves survivability. The player therefore becomes more capable both through inherited traits and through learning the behavior of the ecosystem.

Part of the goal of *Cambrian* is to embrace a design philosophy that prioritizes the sandbox. Emergent and captivating gameplay is driven by interactive elements, and progression that expands utility, physics, and player actions contributes more to that sandbox than progression based solely on percentage increases. This does not mean that percentage-based upgrades are inherently poor design; they can contribute meaningfully when tied to an interactive system and used purposefully. The Improved Tail evolution in *Cambrian* is an example. It is largely a percentage-based mobility increase, but that increase acts directly upon the game's movement physics, changing the player's ability to explore, pursue, escape, and control engagements. I will die on this hill of design outside the context of tabletop-style RPGs.

This is an agency-driven systemic design philosophy: I call it "Possibility-Driven" or "Toolbox design" and it goes beyond sandbox games and progression and into all aspects of a game, for example the difference between a fun open world and a w key simulator.

2. Genre

Cambrian is a 2D survival-action game with roguelite and evolutionary progression elements. Its moment to moment play combines exploration, resource collection, predator avoidance, maneuver based combat, and risk management. Its longer term structure resembles a roguelite: an individual life ends, but successfully preserved resources are converted into permanent lineage level capabilities that affect future generations.

3. Concept and Background

The project interprets the assigned underwater theme through a prehistoric marine ecosystem. Rather than treating the setting as decoration, the game uses predation, feeding, reproduction, death, inheritance, and adaptation as the basis of its rules. The player begins as a comparatively vulnerable organism and gradually develops traits that open new survival strategies.

Cambrian is not intended to be a literal simulation of evolutionary biology. Its evolutionary concepts are gamified so that adaptation can be experienced through player choice, mechanical change, and repeated generations.

4. What the Game Teaches

The prototype teaches its systems through repeated interaction rather than a separate tutorial level. Early play teaches basic locomotion, feeding, predator avoidance, mating, banking EP, and the consequence of death. Later generations introduce inherited adaptations that alter the player's available strategies.

- Movement: aim the organism and propel it through the water while managing momentum and turning.
- Feeding: consume plankton to accumulate Evolution Points.
- Risk management: unpreserved progress can be lost, encouraging the player to decide when to seek a mate.
- Reproduction: reaching a compatible mate allows current progress to be banked for the lineage.
- Predator behavior: predators detect, chase, bite, retreat, and expose positional vulnerabilities that skilled players can exploit.
- Combat: jaws allow predators to be damaged and killed, but direct fighting remains dangerous.
- Evolution: preserved EP is spent after death on inherited adaptations that change future play.
- Spatial awareness: sensory organs provide directional warning of nearby predators.

5. Core Gameplay Loop

1. Spawn as the current generation of the lineage.
2. Explore the underwater environment and consume plankton for EP.
3. Avoid predators or, after evolving jaws, choose to fight them.
4. Search for a compatible mate and reproduce to preserve accumulated EP.
5. Continue exploring and decide how much additional risk to take before death.
6. On death, enter the Generation End evolution screen.
7. Spend preserved EP on inherited adaptations.
8. Begin the next generation with the purchased traits and repeat the loop.

This loop creates a tension between greed and preservation. Remaining alive longer can produce more EP, but unbanked progress is vulnerable. Reproduction acts as a checkpoint for the lineage rather than a conventional save point.

6. Controls (controller recommended)

On a keyboard it is recommended to place the left hand on the conventional WASD with the thumb on the Space Bar and right hand on the arrow keys.

Action	Keyboard	Controller
Aim / Direction	Arrow keys	Directional input
Propel	W	Right Trigger
Reproduce	E	Y
Bite	SPACE	A

7. Rules and Core Mechanics

Movement

The player aims independently of propulsion. Without paired fins, the fish gradually turns toward the requested direction, producing curved movement. Paired fins change this rule by allowing immediate directional positioning. The improved tail increases propulsion and maximum speed. (see above for why this is within toolbox design)

Feeding and Evolution Points

Plankton acts as the primary low-risk resource. Entering the feeding area consumes plankton and awards EP. This creates a dependable progression path while allowing more dangerous activities to coexist as optional risks.

Reproduction and Banking

Compatible mates are limited resources in the world. When the player enters mating range and reproduces, the current lineage progress is preserved. A used mate despawns, preventing repeated banking from the same individual and forcing continued exploration. Spawn-rate tuning makes mates less common than plankton so that preservation requires intentional movement through the environment.

Predators and Combat

Predators wander while idle and pursue the player after detection. Their frontal attack region is intentionally dangerous, while vulnerable spots above and below reward maneuvering. Jaws allow the player to damage predators, but combat remains costly enough that predators retain their role as threats rather than becoming simple targets. Do not fight all of them, you will die.

Health and Death

Predator attacks reduce health. Dermal armor reduces incoming damage. When health reaches zero, the current generation ends, active play stops, and the evolution screen becomes the transition into the next generation.

Sensory Organs

The sensory upgrade activates a circular detection area around the player. When a predator enters this range, the HUD displays a directional warning at the edge of the screen. This provides information rather than direct power: the player must still decide whether to flee, maneuver around the threat, or engage it. Only one predator in range will be tracked at a time to keep the player on their toes.

8. Evolution System

Adaptation	Mechanical Effect	Design Purpose
Jaws	Unlocks biting and predator damage.	Turns combat from impossible into an optional high risk strategy.
Paired Fins	Enables immediate directional changes.	Rewards maneuvering and makes predator blind spot play more controllable.
Improved Tail	Improves propulsion and maximum speed.	Strengthens pursuit, escape, and exploration mobility.
Sensory Organs	Provides directional predator warnings.	Adds information and earlier threat awareness.

Dermal Armor	Reduces incoming damage.	Makes mistakes less punishing without removing predator danger.
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A key design decision was to avoid making every adaptation a percentage increase (prioritizing interaction based sandbox design). Several upgrades instead alter the player's abilities and allow them to do new actions. Jaws introduce an action that did not previously exist; paired fins change the movement model; and sensory organs reveal information that an unevolved organism cannot access.

9. Generational Level Structure

Cambrian treats lives/generations as its level structure (the roguelite elements). A generation has a clear beginning, a period of active play, an ending condition through death, and an interstitial progression screen before the next generation. This structure connects the level transition directly to the subject of inheritance.

Generation 1 — Primitive Survival

The player learns propulsion, feeding, mating, preservation of EP, predator avoidance, health, and death while operating with the lineage's primitive capabilities.

Generation 2 — Adaptation

The next generation inherits adaptations purchased after the previous death. The player must learn how these new capabilities change previously encountered situations—for example, using paired fins to redirect instantly or jaws to turn a predator encounter into a possible combat engagement.

Generation 3 and Beyond — Selection

Later generations combine previously learned mechanics. The player can use sensory information to anticipate a predator, maneuver around its frontal attack region, use mobility traits to control distance, and decide whether the health cost of combat is worthwhile. In this sense, progression is both mechanical inheritance and accumulated player skill.

10. Game Design Schemas: Salen and Zimmerman

The design analysis uses Katie Salen Tekinbaş and Eric Zimmerman's Rules of Play framework. Their work presents game design through conceptual schemas and emphasizes games as interactive systems. For this report, Cambrian is examined through the broad lenses of rules, play, and culture, with particular attention to meaningful play and emergence.

Rules — The Formal System

Cambrian's formal rules establish a compact system of resources, threats, and inheritance. Plankton grants EP; reproduction preserves progress; death ends a generation; preserved EP purchases traits; predators create health pressure; and traits modify future interactions. These rules create a possibility space in which the player chooses between continued resource gathering, preservation, evasion, and combat.

Play — The Experienced System

The player's experience emerges from interacting with those rules. A predator is initially something to avoid. After jaws evolve, the same creature becomes both a threat and a possible opponent. After paired fins evolve, the player can exploit positional openings more precisely. Sensory organs change

anticipation by revealing threat direction before direct contact. The meaning of the same world therefore changes as the lineage develops.

Culture — Context and Representation

The Cambrian-inspired setting contextualizes the mechanics through an evolutionary theme. Feeding, predation, reproduction, inherited traits, and survival pressure give thematic meaning to the formal systems. The game uses these concepts as stylized game abstractions rather than presenting itself as a scientific simulation.

Meaningful Play

The reproduction system is a central example of meaningful play because the consequence of the player's choice is immediately discernible and remains integrated into later generations. Reproducing changes EP from temporary progress into preserved lineage progress. Likewise, purchasing an adaptation produces a visible and mechanical consequence during the next life rather than functioning only as an abstract statistic.

Emergence

The prototype uses relatively simple rules to create unscripted situations. A wandering predator may intersect the player while the player is searching for a mate; the player may choose to abandon food to preserve health, fight after evolving jaws, or use superior maneuverability to pass through a safer approach. These encounters are not individually scripted. They emerge from movement, detection, hitboxes, resource placement, spawning, and player choice.

11. Development and Iteration

The prototype was developed iteratively. Each major revision either added a required piece of the core loop or responded to behavior observed during playtesting.

Iteration	Change and Rationale
Movement prototype	Implemented directional aiming, propulsion, drag, camera following, and constrained pitch to establish underwater locomotion.
Feeding and EP	Added plankton consumption and Evolution Points, creating the first repeatable resource-gathering objective.
Reproduction	Added compatible mates, mating range, and EP banking so progression could be preserved before death.
Shared FishBody	Moved visual orientation and inherited anatomy into a reusable FishBody scene so the player and mates could share lineage appearance.
Predator prototype	Added detection, pursuit, attacks, health damage, retreat behavior, and player death.
Evolution screen	Added a generation-end interface for spending preserved EP and beginning the next generation.
Jaws and combat	Made jaws visible and functional, added a bite hitbox, and allowed predators to take damage and die.
Directional predator combat	Refined predator attack and body hitboxes so frontal engagement is dangerous while skilled players can weave around vulnerable upper/lower approaches.

Paired fins	Changed fins from a minor turning modifier into immediate directional control, producing a clear qualitative movement upgrade.
Sensory organs	Added a large sensory area and HUD threat indicator that communicates predator direction.
Ecosystem motion	Added wandering behavior to mates and predators so the environment feels active rather than populated by stationary test objects.
Population spawning	Added mate and predator spawning to support exploration beyond hand-placed actors.
Balance pass	Expanded world size and predator detection range, increased evolution costs, and reduced mate spawn frequency to make exploration, survival, and reproduction more consequential.

12. Playtesting and Design Decisions

Playtesting exposed several issues that directly changed the design. The first iteration of movement felt less like swimming and more like ice. The organism rotated 360 degrees fully and would not be upright unless facing to the right. This inspired the change of swimming to flip the organism based on direction and cap vertical facing. After orientation flipping was implemented early sprite and collision orientation logic worked when facing right but mirrored incorrectly when facing left, so visual orientation, feeding areas, bite areas, and collision transforms were separated and corrected. There was also the early problem of orientating evolved parts of the fish as well as mates matching the player's look. To solve this, the sprite work was moved to a new node that both the player and mate use called "fish body", allowing evolution to be as simple as rig the new part on the fish body, and set its visibility to true when the evolution is bought. It also ensured the mate would match the player. Predator contact initially behaved more like body blocking than predation; detection, attack areas, retreat behavior, and damage were introduced to create readable attack cycles.

Combat testing also showed that a simple head-on damage race was less interesting than positional play. The predator hitboxes were therefore shaped to leave limited vulnerable approaches above and below the main body, if you weave around the mouth while having jaws you can bite its back or stomach before it repositions. This made paired fins especially valuable, which motivated changing that upgrade from a modest turning multiplier to immediate directional redirection.

Later ecosystem testing motivated a larger world, longer predator detection, higher evolution costs, and lower mate spawn rates. These changes reduce the frequency of effortless reproduction and make exploration and survival decisions more important to the generational loop. Early iterations allowed one to just farm to 50 points quickly, mate, die then buy all evolutions at once. It was simply too easy.

I found iteration useful not in just the gameplay design, but in encouraging an architecture that scales better as one adds to it. Iteration quickly brought me to make solutions that make the iterations quicker and easier. And when the game has a base sprite with 5 other sprites progressively plastered on it. That becomes important to prevent headaches in building the game in the long run.

13. Current Prototype State

- Player swimming, propulsion, drag, camera following, and directional orientation are functional.
- Plankton feeding and EP accumulation are functional.

- Mates wander, detect reproduction range, bank progress, and despawn after use.
- Predators wander, detect and chase the player, attack, retreat, take bite damage, and can die.
- Health, damage, death, and generation-end flow are functional.
- All five planned adaptations have mechanical hooks: jaws, paired fins, improved tail, sensory organs, and dermal armor.
- The sensory HUD warning is functional.
- Predator and mate population spawning is functional.
- Generation-based progression provides the prototype's level structure.

14. Conclusion

Cambrian uses a compact set of underwater survival mechanics to make generational adaptation the central structure of play. Its strongest design relationship is between player learning and lineage evolution: later generations are more capable because the organism has inherited new traits, while the human player has simultaneously learned how to read predators, manage risk, navigate the environment, and preserve progress. The resulting prototype demonstrates a complete loop of forage, survive, reproduce, die, evolve, and begin again.

References

Salen Tekinbaş, Katie, and Eric Zimmerman. Rules of Play: Game Design Fundamentals. MIT Press, 2003.

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